National University

Of Computer & Emerging Sciences Chiniot- Faisalabad Campus

CS 1002 – Programming Fundamentals (Fall 2026)

Assignment 3

Topics Covered:	**Submission Deadline:** *Wednesday – October 28, 2026 by 23.59 sharp*
▪ C++ loops, nested loops ▪ Random numbers	

Submission Guidelines:

- This is an individual assignment.
- Use of any advanced topics is not acceptable (e.g., arrays, strings, etc.).
- Solve each question in a separate .cpp file.
- Create a word file with C++ code and screenshot of the output for each question.
- Put all the files in a folder and compress as .zip for submission.

Problem # 1

Write a C++ program that generates a random math quiz for the user. The program should use a for loop to present a series of math questions. The user should input answers, and the program should provide feedback on whether each answer is correct or incorrect. Use switch structure for selection.

a. The program should ask the user for the number of math questions to generate.
b. Using a `for` loop, generate the specified number of random math questions. Each question
c. should involve two random integers between 1 and 10 and one random arithmetic operator (+, -, *, or /).
d. Present each question to the user in the following format: "Question X: Y [operator] Z = ?" where X is the question number, Y and Z are the random integers, and [operator] is the random operator.
e. Prompt the user to enter the answer for each question.
f. Check if the user's answer is correct, and provide feedback. If the answer is correct, display "Correct!" for that question. If the answer is incorrect, display "Incorrect. The correct answer is [correct answer]."

g. At the end of the quiz, display the user's score as the number of correct answers out of the total number of questions.

Sample run:

How many math questions do you want to generate? **5**

Question 1: 7 + 4 = ?

Enter your answer: **11**

Correct!

Question 2: 9 * 2 = ?

Enter your answer: **18**

Correct!

Question 3: 1 - 3 = ?

Enter your answer: **2**

Incorrect. The correct answer is -2.

Question 4: 6 / 3 = ?

Enter your answer: **2**

Correct!

Question 5: 10 - 7 = ?

Enter your answer: **3**

Correct!

Your score = 4 out of 5

Problem # 2

Write a C++ program that inputs a positive integer 'n' and generates and solves the following series:

$$\frac{1}{2}+\frac{2}{3}+\frac{3}{4}+\frac{4}{5}+\cdots+\frac{n}{n+1}$$

Your program should:

a. Display all 'n' terms in fractional form (see the example below).
b. Calculate the sum of the series using floating-point arithmetic and displays the answer upto four decimal places.

Enter n: **4**
Output:
1/2 + 2/3 + 3/4 + 4/5
Sum: 2.7167

Problem # 3

Write a C++ program that inputs a positive integer \(n\) and generates the following series for 'n' terms:

$$1^2 - 2^2 + 3^2 - 4^2 + 5^2 - \cdots$$

Your program should:

a. Display the first 'n' terms with alternating positive and negative signs.
b. Calculate and display the sum of the series.

Enter n: **5**
Output:
1 - 4 + 9 - 16 + 25
Sum: 15

Problem # 4

Write a C++ program that inputs two positive integers num1 and num2, where num1 < num2, and performs the following tasks.

a. Take valid inputs from the user, num1 < num2.
b. Display all prime numbers between num1 and num2, both inclusive.
c. Count the total number of prime numbers in the range.
d. Find the largest gap between two consecutive prime numbers in the range.
e. Display the pair of consecutive primes that produces the largest gap.
f. Identify all pairs of twin primes in the range. Twin primes are prime numbers that differ by exactly 2, such as (11,13).

Enter num1 and num2 (num1 < num2): **51 30**
Error, renter the values.

Enter num1 and num2 (num1 < num2): **10 30**

Prime numbers: 11 13 17 19 23 29
Total prime numbers: 6
Largest gap: 6
Prime pair: 23 and 29
Twin primes: (11, 13); (17, 19);

Problem # 5

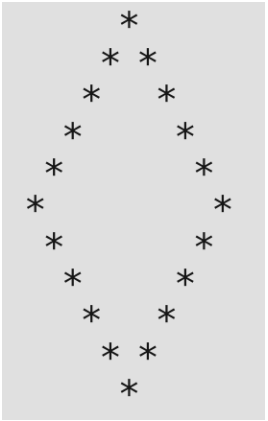
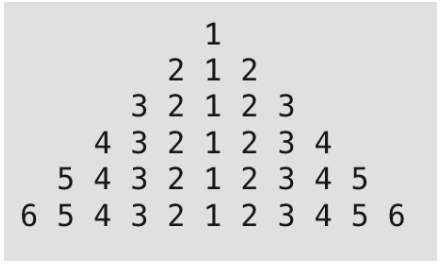
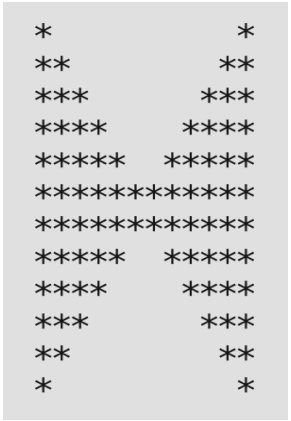
Write a C++ program that find number spikes in 3 rounds. In each round the user enters positive integers, one at a time and will terminate by entering a negative value. From the 3rd number onward in a round, compare the current number with the sum of the previous two numbers entered in that round. If the current number is strictly greater than that sum, display a spike message and count it. After each round, display that round's spike count. After all rounds, display the total spikes. The previous two numbers must not carry over from one round to the next.

Sample input:
Enter Round 1 values (negative value to terminate): 2 3 4 10 5 20 -1
Enter Round 2 values (negative value to terminate): 100 19 3 1 -99
Enter Round 3 values (negative value to terminate): 9 0 1 30 6 2 -771

Sample output:
Spike at position 4: 10 > 3 + 4
Spike at position 6: 20 > 10 + 5
Round 1 spikes: 2
Round 2 spikes: 0
Spike at position 4: 30 > 0 + 1
Round 3 spikes: 1
Total spikes: 3

Problem # 6

Write a C++ program to generate the following patters. Provide a menu-driven interface (use do-while loop) to print a desired pattern. The user will exit the program on entering a ZERO.

Hollow diamond pyramid	Palindrome traingle pattern	Butterfly star pattern
Enter N: 6	Enter N: 6	Enter N: 6
		

Problem # 7

Write a C++ program that generates a funny and unpredictable story by randomly selecting words from different categories, such as characters, places, actions, and objects. The program should generate N sentences, see the sample run below.

1. Each sentence must contain the following 4 types of words, selected randomly:
    - Character: student, teacher, robot, alien
    - Action: discovered, chased, stole, carried
    - Place: classroom, moon, cafeteria, jungle
    - Object: laptop, pizza, treasure, dinosaur
2. Since arrays and strings are not allowed, use switch statements to select and display each word randomly from each category. Use ONLY one switch structure for all words types.
3. Display the selected words in the following order to form each sentence:

Character + Action + Object + Place

4. Begin each sentence with its sentence number and end it with a full stop.
5. You can add appropriate prepositions with each place instead of a single word (e.g., write 'on the moon' instead of 'moon').

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RANDOM STORY GENERATOR
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How many sentences do you want? 3

Sentence 1:
The alien discovered a pizza in the jungle.

Sentence 2:
The teacher chased a dinosaur in the cafeteria.

Sentence 3:
The student stole a laptop on the moon.
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